Bounded Independence for k-Min-Wise Hashing: Tight Bounds and Limitations of Structured Hashing

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Abstract

Min-wise hashing and its k-min-wise extension are fundamental tools in sampling, sketching, similarity estimation, etc. A standard approach to constructing such families is bounded independence. For ordinary min-wise hashing, the required degree of independence is fully understood: $\Theta(\log 1/\delta)$-wise independence is both sufficient and necessary. For k-min-wise hashing, however, the best previous result only showed that $O(k \log\log 1/\delta + \log 1/\delta)$-wise independence suffices, with no matching lower bound.

We give a tight characterization of the amount of bounded independence required for k-min-wise hashing, proving that $\Theta(k + \log 1/\delta)$-wise independence is both sufficient and necessary. This improves the previous upper bound and provides a matching lower bound. Consequently, the standard construction of bounded-independent hash families has seed length $O\big((k+\log 1/\delta)\cdot \log(N/\delta)\big)$. In particular, for polynomially small δ and any $\Omega(\log N) \le k \le N^{1-c}$, it achieves the optimal seed length $O(k \log N)$.

We further investigate two standard low-independence hash families. For random affine functions over $\mathbb{F}_2$, which form a pairwise independent family, we show that the multiplicative error is $\Omega(\log N)$ even for ordinary min-wise hashing. For simple tabulation hashing, which is 3-wise independent and performs well for ordinary min-wise hashing, we show that it incurs a multiplicative error $\Omega(N)$ for k-min-wise hashing whenever $k \ge 4$.

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1 Introduction

Min-wise hashing, introduced by Broder, Charikar, Frieze, and Mitzenmacher [BCFM98], is a fundamental primitive in randomized algorithms, with applications to similarity and rarity estimation, near-duplicate detection, and ℓ_0-sampling in streaming algorithms [CDF+01, DM02, Hen06, CF14]. Intuitively, min-wise hashing uses a hash function to approximately simulate a uniformly random ordering: for any designated element, the probability that it receives the smallest hash value should be close to that under a random permutation. Its k-min-wise extension, introduced by Feigenblat, Porat, and Shiftan [FPS11], requires an analogous guarantee for any prescribed set of at most k elements: the probability that all of them receive smaller hash values than all remaining elements should be close to the corresponding probability under a uniformly random permutation.

Definition 1.1 (k-Min-Wise Hashing)**.** *We use $a = b \pm \delta$ to denote $a \in [b-\delta, b+\delta]$. Let $0 < \delta < 1$, and let $\mathcal{H}$ be a family of functions from $[N]$ to an ordered range $[M]$. We say that $\mathcal{H}$ is a k-min-wise hash family with multiplicative error δ if for every pair of sets X and $Y \subseteq [N]$ satisfying $X \cap Y = \varnothing$ and $|Y| \leq k$,*

$$\Pr_{h\sim\mathcal{H}}\left[\max_{y\in Y} h(y) < \min_{x\in X} h(x)\right] = (1\pm\delta)\cdot\binom{|X|+|Y|}{|Y|}^{-1}.$$

The case $k = 1$ is called min-wise hashing.

Under a fully random continuous ordering, the relative ordering of $X \cup Y$ is a uniformly random permutation. Hence every $|Y|$-subset is equally likely to occupy the first $|Y|$ positions, and the probability in the definition is exactly $\binom{|X|+|Y|}{|Y|}^{-1}$. For a finite range, ties between the elements of X and Y introduce a small additional error. Throughout the paper, we take $M = \Omega(N/\delta)$, for which a fully random function into $[M]$ already satisfies the definition within an $O(\delta)$ multiplicative error, as stated by Indyk [Ind01].

A central goal of hashing is to construct such families using as little randomness as possible. One of the most natural approaches is bounded independence. A t-wise independent hash family agrees exactly with a fully random function on every collection of at most t inputs and admits the standard polynomial construction with seed length essentially linear in t. This leads to the question of determining the amount of independence needed for k-min-wise hashing as a function of k and δ.

1.1 Prior Works

For ordinary min-wise hashing, Indyk [Ind01] proved that $O(\log 1/\delta)$-wise independence suffices to achieve multiplicative error δ. Pătraşcu and Thorup [PT16] proved a matching lower bound. Thus, for min-wise hashing, the power of bounded independence is completely characterized: the optimal degree is $\Theta(\log 1/\delta)$.

For k-min-wise hashing, Feigenblat, Porat, and Shiftan [FPS11] proved that $O(k \log\log 1/\delta + \log 1/\delta)$-wise independence suffices. No matching lower bound depending on k was known. In particular, for $k = \Theta(\log N)$ and $\delta = N^{-c}$, their analysis combined with the standard construction of bounded independence (e.g. [ABI86]) gives seed length $O(k \log N \log\log N)$.

There is also a line of work constructing min-wise hash families using general pseudorandomness tools. Saks, Srinivasan, Zhou, and Zuckerman [SSZZ00] related min-wise hashing to pseudorandom generators for combinatorial rectangles, and Gopalan and Yehudayoff [GY20] later improved the seed length. Building on pseudorandom generators for combinatorial rectangles together with additional pseudorandomness techniques, Chen, Huang, and Li [CHL26] recently constructed explicit k-min-wise families using $O(k \log N)$ random bits and achieving multiplicative error $2^{-O(\log N/\log\log N)}$

for $k = \log^{O(1)} N$. Their seed length is optimal up to constant factors, while the error is almost polynomially small. This raises the question of whether one can simultaneously obtain the optimal $O(k \log N)$ seed length and polynomially small error in the large-k regime.

Besides general constructions, another line of work studies the min-wise behavior of concrete hash families. Two particularly classical examples are modular affine hashing and tabulation hashing, both belonging to the universal-hashing framework initiated by Carter and Wegman [CW79, WC81]. These families are simple to represent and evaluate and have been extensively studied in hash tables and randomized data structures. The one-dimensional modular affine family $h(x) = (ax + b) \bmod p$ is a canonical pairwise-independent construction. Nevertheless, Broder, Charikar, Frieze, and Mitzenmacher [BCFM98] and Pătraşcu and Thorup [PT16] showed that it incurs multiplicative error $\Omega(\log N)$ for ordinary min-wise hashing.

For simple tabulation hashing, the universe $[N]$ is viewed as Σ^c, where $|\Sigma| = N^{1/c}$, and a key $x = (x_1, \ldots, x_c)$ is hashed as $h(x) = \bigoplus_{i=1}^{c} T_i[x_i]$ using independent random tables. Although simple tabulation is only 3-wise independent, Pătraşcu and Thorup [PT12] showed that it provides unexpectedly strong guarantees for several classical applications. In particular, for a set of n keys, they proved a multiplicative min-wise error of $O\left(\frac{\log^2 N}{N^{1/c}}\right)$. Pătraşcu and Thorup [PT13] subsequently developed a cleaner related analysis for twisted tabulation and obtained the same error bound. These results motivate asking whether the favorable min-wise behavior of structured low-independence families extends to the stronger k-min-wise property.

1.2 Our Results

We completely characterize the amount of bounded independence required for k-min-wise hashing.

Theorem 1.2 (Informal Version of Results in Section 2)**.** *The degree of independence necessary and sufficient to guarantee k-min-wise hashing with multiplicative error δ is*

$$\Theta\left(k + \log \frac{1}{\delta}\right).$$

The upper bound improves the result of Feigenblat, Porat, and Shiftan [FPS11] by removing the additional $\log \log 1/\delta$ factor multiplying k. We also provide a matching lower bound, proving that both the dependence on k and the dependence on $\log 1/\delta$ are necessary. This gives a tight characterization of the power of bounded independence for constructing k-min-wise hash families.

The standard construction of bounded independence consequently has seed length $O\big((k + \log 1/\delta) \cdot \log(N/\delta)\big)$. In particular, when $\delta = N^{-c}$ and $k = \Omega(\log N)$, this becomes $O(k \log N)$. In the polylogarithmic-k regime, this seed length is optimal up to constant (see Appendix A). Hence, for $\Omega(\log N) \le k \le \log^{O(1)} N$, we improve the almost-polynomial error of [CHL26] to polynomially small error while preserving the optimal seed length. More generally, our bounded-independence characterization applies to every k and continues to yield polynomially small error with seed length $O(k \log N)$.

Beyond the generic bounded-independence construction, we study two standard low-independence hash families and show that their additional structure does not lead to stronger k-min-wise guarantees.

We first consider random affine functions over $\mathbb{F}_2$, which are the natural higher-dimensional analogue of the classical modular affine family $h(x) = (ax + b) \bmod p$. The latter is known to incur multiplicative error $\Omega(\log N)$ for ordinary min-wise hashing [BCFM98, PT16]. We prove that the same limitation persists over binary vector spaces.

Theorem 1.3 (Informal Version of Theorem 3.1)**.** *There exists a set S of size N such that, for a uniformly random affine function $h(x) = Ax \oplus b$,*

$$\Pr\left[h(0) < \min_{s \in S} h(s)\right] = \Omega\left(\frac{\log N}{N}\right).$$

Consequently, random affine hashing incurs multiplicative error $\Omega(\log N)$ for min-wise hashing.

We next consider simple tabulation hashing. Pătraşcu and Thorup [PT12] showed that, despite being only 3-wise independent, simple tabulation provides strong guarantees for ordinary min-wise hashing, with error only $O\left(\frac{\log^2 N}{N^{1/c}}\right)$. We show that this favorable behavior does not extend to k-min-wise hashing once $k \geq 4$.

Theorem 1.4 (Informal Version of Theorem 4.1)**.** *For every sufficiently large N, there exist disjoint sets X and Y with $|X| = N$ and $|Y| = 4$ such that simple tabulation hashing satisfies*

$$\Pr\left[\max_{y \in Y} h(y) < \min_{x \in X} h(x)\right] = \Omega\left(\frac{1}{N^3}\right).$$

Since the corresponding probability under a uniformly random ordering is $\Theta(1/N^4)$, simple tabulation incurs multiplicative error $\Omega(N)$ for 4-min-wise hashing, and therefore for every $k \geq 4$.

1.3 Relation to the Concurrent Preprints

This paper merges and extends two concurrent and independent preprints: Chen, Huang, and Li, *Constructions of k-Min-Wise Hash from Bounded Independence* (arXiv:2607.27157v1), and Wang, *Limited Independence Suffices for Large-k Min-wise Hashing* (arXiv:2607.10255v2). Both preprints independently established the $O(k + \log 1/\delta)$ upper bound using different arguments. The present joint version gives a unified presentation and additionally includes the matching lower bound and the results on random affine hashing and simple tabulation hashing.

1.4 Notation and Basic Definitions

Let $[n]$ denote $\{1, 2, \ldots, n\}$ and $\binom{n}{k}$ denote the binomial coefficient. Unless stated otherwise, all logarithms are to base 2. For a finite family Ω, we use $x \sim \Omega$ to denote that x is sampled uniformly from Ω. For three real variables a, b and δ, $a = b \pm \delta$ means the error between a and b is $|a - b| \leq \delta$.

For a hash function h and a set S, let

$$h(S) := \big(h(s) : s \in S\big)$$

denote the *multiset* of hash values of the elements of S. For every nonempty set S, we use the shorthand

$$\min h(S) := \min_{s \in S} h(s), \qquad \max h(S) := \max_{s \in S} h(s).$$

We recall the definition of bounded-independent hash families.

Definition 1.5 (t-wise Independent Hash Family [CW79, WC81])**.** *A hash family $\mathcal{H}$ from a domain $\mathcal{U}$ to a range $\mathcal{R}$ is called t-wise independent if, for any distinct inputs $x_1, \ldots, x_t \in \mathcal{U}$ and any outputs $y_1, \ldots, y_t \in \mathcal{R}$,*

$$\Pr_{h \sim \mathcal{H}}\big[h(x_1) = y_1, \ldots, h(x_t) = y_t\big] = |\mathcal{R}|^{-t}.$$

Explicit constructions of t-wise independent hash family require $\Theta\big(t \cdot (\log |\mathcal{U}| + \log |\mathcal{R}|)\big)$ random bits [ABI86].

2 Bounded Independence

We now determine the amount of bounded independence needed to guarantee k-min-wise hashing. For the upper bound, we fix disjoint sets X and Y and enumerate the possible values of $\theta := \max h(Y)$. This expresses $\Pr[\max h(Y) < \min h(X)]$ as a weighted sum of the probabilities $\Pr[\min h(X) > \theta]$. We divide the sum according to the size of θ. For small θ, truncated inclusion–exclusion shows that $\Pr[\min h(X) > \theta]$ under bounded independence is close to the corresponding probability under fully random hashing. For large θ, a tail bound for limited independence shows that the total contribution of this range is already small under both distributions. Combining the two estimates leaves only a small additive discrepancy, which is sufficiently small relative to the ideal probability $\binom{|X|+|Y|}{|Y|}^{-1}$ to yield the desired multiplicative error.

For the lower bound, we impose a parity constraint on the number of elements of Y whose hash values fall in one half of the range. This preserves $(k-1)$-wise independence but biases the probability of the event $\max h(Y) < \min h(X)$, proving that $\Omega(k)$-wise independence is necessary. The additional $\Omega(\log 1/\delta)$ lower bound follows from the min-wise lower bound of Pătraşcu and Thorup [PT16]. Together, these two bounds give the matching characterization.

2.1 Upper Bound

We first prove that $O(k+\log 1/\delta)$-wise independence suffices for k-min-wise hashing. This improves the previous upper bound $O(k \log\log 1/\delta + \log 1/\delta)$ of [FPS11].

Theorem 2.1. *There exists a constant $c > 0$ such that, if $M \geq N/\delta$, then every $c\cdot(k+\log 1/\delta)$-wise independent hash family from $[N]$ to $[M]$ is k-min-wise with error 3δ.*

Let $X, Y \subseteq [N]$ be disjoint, let $|X| = s$, and, without loss of generality, consider the case $|Y| = k$. We use $\Pr_\ell$ to denote probability under an ℓ-wise independent hash family, while Pr without a subscript denotes probability under fully random hashing.

We consider $(\ell+k)$-wise independence. Directly analyzing the joint event $\{\max h(Y) < \min h(X)\}$ is difficult because the hash values of the elements in X and Y may be correlated. We therefore enumerate $\theta := \max h(Y)$ and obtain

$$\Pr_{\ell+k}[\max h(Y) < \min h(X)] = \sum_{\theta=1}^{M} \frac{\theta^k - (\theta-1)^k}{M^k} \cdot \Pr_\ell[A_\theta],$$

$$\Pr[\max h(Y) < \min h(X)] = \sum_{\theta=1}^{M} \frac{\theta^k - (\theta-1)^k}{M^k} \cdot \Pr[A_\theta],$$

where

$$A_\theta = \{\min h(X) > \theta\} \quad \text{and} \quad Z_\theta = \big|h(X) \cap [1,\theta]\big|.$$

Enumerating the hash values of the k elements of Y consumes k degrees of independence, leaving ℓ-wise independence on X.

Set $T = \frac{c\ell M}{s} \in \mathbb{N}$, where $c < 1$ is a constant to be determined, and split the summation into two parts:

$$\Pr_{\ell+k}[\max h(Y) < \min h(X)] = \sum_{\theta \leq T-1} \frac{\theta^k - (\theta-1)^k}{M^k} \Pr_\ell[A_\theta] + \sum_{\theta \geq T} \frac{\theta^k - (\theta-1)^k}{M^k} \Pr_\ell[A_\theta].$$

The First Part. We first consider $\Pr_\ell[A_\theta]$ and use the Bonferroni inequalities.

Lemma 2.2 (Bonferroni Inequalities)**.** *Let $A_1, \ldots, A_n$ be n events. Then, for every $r = 0, 1, \ldots, n$,*

$$\Pr\left[\bigcap_{i=1}^{n} A_i\right] \begin{cases} \leq \displaystyle\sum_{t=0}^{r} (-1)^t \sum_{\mathcal{I}\subseteq[n],\, |\mathcal{I}|=t} \Pr\left[\bigcap_{j\in\mathcal{I}} \overline{A_j}\right], & r \text{ is even,} \\ \geq \displaystyle\sum_{t=0}^{r} (-1)^t \sum_{\mathcal{I}\subseteq[n],\, |\mathcal{I}|=t} \Pr\left[\bigcap_{j\in\mathcal{I}} \overline{A_j}\right], & r \text{ is odd.} \end{cases}$$

Assume that ℓ is even. Then

$$\sum_{j=0}^{\ell-1} (-1)^j \sum_{\substack{X'\subseteq X \\ |X'|=j}} \Pr_\ell[h(X') \subseteq [1,\theta]] \leq \Pr_\ell[A_\theta] \leq \sum_{j=0}^{\ell} (-1)^j \sum_{\substack{X'\subseteq X \\ |X'|=j}} \Pr_\ell[h(X') \subseteq [1,\theta]].$$

Consequently,

$$\Pr_\ell[A_\theta] = \sum_{j=0}^{\ell-2} (-1)^j \sum_{\substack{X'\subseteq X \\ |X'|=j}} \Pr_\ell[h(X') \subseteq [1,\theta]] \pm \Delta,$$

where

$$\Delta \leq \sum_{j=\ell-1}^{\ell} \sum_{\substack{X'\subseteq X \\ |X'|=j}} \Pr_\ell[h(X') \subseteq [1,\theta]] \leq 2\binom{s}{\ell-1}\left(\frac{\theta}{M}\right)^{\ell-1} \leq 2\left(\frac{es\theta}{M(\ell-1)}\right)^{\ell-1}.$$

The above analysis involves only subsets of size at most ℓ. Hence the relevant probabilities are identical under fully random hashing and ℓ-wise independent hashing, and the same argument gives

$$\Pr[A_\theta] = \sum_{j=0}^{\ell-2} (-1)^j \sum_{\substack{X'\subseteq X \\ |X'|=j}} \Pr_\ell[h(X') \subseteq [1,\theta]] \pm \Delta.$$

Combining the two estimates yields

$$\Pr_\ell[A_\theta] = \Pr[A_\theta] \pm 2\Delta = \Pr[A_\theta] \pm 4\left(\frac{es\theta}{M(\ell-1)}\right)^{\ell-1}.$$

Substituting this estimate into the summation, we obtain

$$\begin{aligned} \sum_{\theta\leq T-1} \frac{\theta^k - (\theta-1)^k}{M^k} \Pr_\ell[A_\theta] &= \sum_{\theta\leq T-1} \frac{\theta^k - (\theta-1)^k}{M^k} \left(\Pr[A_\theta] \pm 4\left(\frac{es\theta}{M(\ell-1)}\right)^{\ell-1}\right) \\ &= \sum_{\theta\leq T-1} \frac{\theta^k - (\theta-1)^k}{M^k} \Pr[A_\theta] \pm \frac{4T^k}{M^k}\left(\frac{esT}{M(\ell-1)}\right)^{\ell-1}. \end{aligned}$$

The Second Part. We again consider $\Pr_\ell[A_\theta]$, this time using a concentration inequality. We use the tail bound from [BR94].

Lemma 2.3 (t-wise Independence Tail Bound [BR94, Lemma 2.3])**.** *Let $t \geq 4$ be even, and let $X_1, \dots, X_n$ be t-wise independent random variables taking values in $[0,1]$. Let $X = X_1 + \cdots + X_n$, let $\mu = \mathbb{E}[X]$, and let $A > 0$. Then*

$$\Pr[|X - \mu| \geq A] \leq 8\left(\frac{t\mu + t^2}{A^2}\right)^{t/2}.$$

Let $4 \leq \ell' < \ell$ be an even parameter to be determined. Applying Lemma 2.3 with $t = \ell'$ and $\mu = A = \mathbb{E}[Z_\theta]$, and assuming that $\mathbb{E}[Z_\theta] \geq \ell'$ for every $\theta \geq T$, gives

$$\text{both } \Pr_\ell[A_\theta] \text{ and } \Pr[A_\theta] \leq 8\left(\frac{2\ell' \,\mathbb{E}[Z_\theta]}{(\mathbb{E}[Z_\theta])^2}\right)^{\ell'/2} = 8\left(\frac{2\ell' M}{\theta s}\right)^{\ell'/2}.$$

Substituting this bound into the summation, we obtain

$$\begin{aligned}
\sum_{\theta \geq T} \frac{\theta^k - (\theta-1)^k}{M^k} \Pr_\ell[A_\theta] &\leq \Pr_\ell[A_T] \sum_{\theta \in [T, 2T]} \frac{\theta^k - (\theta-1)^k}{M^k} + \Pr_\ell[A_{2T}] \sum_{\theta \in [2T, 4T]} \frac{\theta^k - (\theta-1)^k}{M^k} + \cdots \\
&\leq 8\left(\frac{2\ell' M}{Ts}\right)^{\ell'/2}\left(\frac{2T}{M}\right)^k + 8\left(\frac{2\ell' M}{2Ts}\right)^{\ell'/2}\left(\frac{4T}{M}\right)^k + \cdots \\
&= 8\left(\frac{2\ell' M}{Ts}\right)^{\ell'/2}\left(\frac{2T}{M}\right)^k \left(1 + \left(\frac{1}{2}\right)^{\ell'/2 - k} + \cdots\right) \\
&\leq 16\left(\frac{2\ell' M}{Ts}\right)^{\ell'/2}\left(\frac{2T}{M}\right)^k.
\end{aligned}$$

The last inequality holds when $\ell'/2 - k \geq 1$, so that the geometric series is at most 2.

The same argument applies under fully random hashing. Therefore,

$$\sum_{\theta \geq T} \frac{\theta^k - (\theta-1)^k}{M^k} \Pr_\ell[A_\theta] = \sum_{\theta \geq T} \frac{\theta^k - (\theta-1)^k}{M^k} \Pr[A_\theta] \pm 32\left(\frac{2\ell' M}{Ts}\right)^{\ell'/2}\left(\frac{2T}{M}\right)^k.$$

Combining the Estimates. Adding the two parts gives

$$\begin{aligned}
\sum_{\theta=1}^{M} \frac{\theta^k - (\theta-1)^k}{M^k} \Pr_\ell[A_\theta] = &\sum_{\theta=1}^{M} \frac{\theta^k - (\theta-1)^k}{M^k} \Pr[A_\theta] \\
&\pm \left(\frac{4T^k}{M^k}\left(\frac{esT}{M(\ell-1)}\right)^{\ell-1} + 32\left(\frac{2\ell' M}{Ts}\right)^{\ell'/2}\left(\frac{2T}{M}\right)^k\right).
\end{aligned}$$

We now choose the parameters. Define an auxiliary error parameter $\delta' = \delta/2^z$, where $z \in \mathbb{N}$ will be determined later.

Let $\ell' = \ell/30$, assuming that $60 \mid \ell$ so that ℓ' is even. Set $c = 1/6$, and hence

$$T = \frac{\ell M}{6s} \in \mathbb{N}.$$

This guarantees that

$$\mathbb{E}[Z_\theta] = \frac{\theta s}{M} \ge \ell' \qquad \text{for every } \theta \ge T.$$

Moreover, assuming $\ell \ge 40$,

$$\frac{esT}{M(\ell-1)} = \frac{e\ell}{6(\ell-1)} < \frac{1}{2}, \qquad \frac{2\ell' M}{Ts} = \frac{2}{5} < \frac{1}{2}.$$

If we further assume that

$$\frac{\ell'}{2} \ge \log\frac{32}{\delta'} \qquad \text{and} \qquad \ell - 1 \ge \log\frac{4}{\delta'},$$

then

$$\text{the total additive error} \le \frac{T^k}{M^k}\delta' + \frac{(2T)^k}{M^k}\delta'.$$

We next choose z and ℓ to satisfy all the constraints above:

$$\begin{cases} 60 \mid \ell, \quad \ell \ge 120; \\ \ell \ge 60k + 60; \\ \ell \ge 60(z + \log 1/\delta + 5). \end{cases} \tag{2.1}$$

We also require

$$\frac{T^k}{M^k}\delta' \le \frac{(2T)^k}{M^k}\cdot\frac{\delta}{2^z} \le \frac{(k/e)^k}{(s+k)^k}\delta \le \frac{\delta}{\binom{s+k}{k}}.$$

It suffices to consider the case $s \ge k$, since the case $s < k$ is handled trivially by $2k$-wise independence. We therefore relax the preceding requirement to

$$\frac{(2T)^k}{M^k}\cdot\frac{\delta}{2^z} \le \frac{(k/e)^k}{(2s)^k}\delta.$$

Substituting $T = \ell M/(6s)$, this is equivalent to

$$\left(\frac{2e}{3}\right)^k \ell^k \le k^k 2^z.$$

Taking logarithms gives

$$z \ge k\log\frac{2e}{3} + k\log\frac{\ell}{k}. \tag{2.2}$$

Let $L = \lceil \log 1/\delta \rceil$. We choose $z = 15k + L + 5$ and $\ell = 60(z + L + 5)$. Thus $\ell = 900k + 120L + 600$, which clearly satisfies all the constraints in (2.1).

To verify (2.2), we use $\log(1+x) \le 1.45x$ for $x \ge 0$:

$$\begin{aligned} k\log\left(\frac{\ell}{k}\right) &= k\log\left(900\left(1 + \frac{120L+600}{900k}\right)\right) \\ &= k\log 900 + k\log\left(1 + \frac{120L+600}{900k}\right) \\ &\le 9.82k + 1.45k\left(\frac{120L+600}{900k}\right) \\ &\le 9.82k + 0.194L + 0.968. \end{aligned}$$

Adding the constant term $k\log(2e/3) \le 0.86k$, the right-hand side of (2.2) is at most

$$0.86k + 9.82k + 0.194L + 0.968 = 10.68k + 0.194L + 0.968.$$

This is smaller than our choice $z = 15k + L + 5$, so (2.2) holds.

Conclusion. We have shown that the muliplicative error of $\Pr_{\ell+k}[\max h(Y) < \min h(X)]$ relative to $\Pr[\max h(Y) < \min h(X)]$ is at most 2δ. It remains to show that a fully random hash function already satisfies the k-min-wise property with error δ when $M \geq N/\delta$, as observed by Indyk [Ind01]. The following is the formal proof.

Lemma 2.4. *Let $X, Y \subseteq [N]$ be disjoint sets with $|X| = s$ and $|Y| = k$, and let $\sigma : [N] \to [M]$ be a fully random function. Then*

$$\frac{1}{\binom{s+k}{k}}\left(1 - \frac{s+k}{M}\right) \leq \Pr_{\sigma}[\max \sigma(Y) < \min \sigma(X)] \leq \frac{1}{\binom{s+k}{k}}.$$

In particular, since $s + k \leq N$, the multiplicative error is at most δ whenever $M \geq N/\delta$.

Proof. The exact probability $P = \Pr_{\sigma}[\max \sigma(Y) < \min \sigma(X)]$ is

$$\begin{aligned} P &= \sum_{\theta=1}^{M} \frac{\theta^k - (\theta-1)^k}{M^k}\left(1 - \frac{\theta}{M}\right)^s \\ &= \sum_{\theta=1}^{M} \int_{(\theta-1)/M}^{\theta/M} kx^{k-1}\left(1 - \frac{\theta}{M}\right)^s \mathrm{d}x. \end{aligned}$$

The ideal probability in the absence of collisions is given by the Beta integral

$$I = \int_0^1 kx^{k-1}(1-x)^s \mathrm{d}x = \frac{1}{\binom{s+k}{k}}.$$

For every $x \in \big[(\theta-1)/M, \theta/M\big]$, we have $1 - \theta/M \leq 1 - x$, and hence

$$P \leq \sum_{\theta=1}^{M} \int_{(\theta-1)/M}^{\theta/M} kx^{k-1}(1-x)^s \mathrm{d}x = I.$$

For the lower bound, we use $1 - \theta/M \geq 1 - x - 1/M$ and convexity:

$$\left(1 - \frac{\theta}{M}\right)^s \geq (1-x)^s - \frac{s}{M}(1-x)^{s-1}.$$

Integrating over all intervals gives

$$P \geq I - \frac{s}{M}\int_0^1 kx^{k-1}(1-x)^{s-1}\mathrm{d}x = I - \frac{s}{M}\frac{1}{\binom{s+k-1}{k}} = I\left(1 - \frac{s+k}{M}\right).$$

Therefore,

$$I\left(1 - \frac{s+k}{M}\right) \leq P \leq I.$$

Thus the multiplicative error is at most $(s+k)/M$. Since $s + k \leq N$, it is at most δ whenever $M \geq N/\delta$. $\square$

Combining the two parts, the multiplicative error relative to $\binom{s+k}{k}^{-1}$ is at most 3δ, while $\ell + k = \Theta(k + \log 1/\delta)$. This proves Theorem 2.1.

An Alternative Argument via Subset Restriction. The bounded-independence argument above handles the second range by applying the tail inequality at a reduced moment order $\ell' < \ell$. We also observe that the intermediate range can be controlled in a different way, without reducing the moment order, by restricting attention to a suitably chosen subset of the competing set. Write $p_\theta := \frac{\theta}{M}$ and $\mu_\theta := sp_\theta = \mathbb{E}[Z_\theta]$.

For a subset $I \subseteq X$ of size b, let $A_\theta(I) := \{\min h(I) > \theta\}$. The Bonferroni calculation above applies equally to I and gives

$$\left|\Pr_\ell[A_\theta(I)] - \Pr[A_\theta(I)]\right| \le 4\left(\frac{ebp_\theta}{\ell-1}\right)^{\ell-1}. \tag{2.3}$$

Thus the error is controlled by the effective mean bp_θ, rather than by the full size s.

For small thresholds, say $\mu_\theta \le \alpha(\ell-1)$ for a sufficiently small constant $\alpha > 0$, we simply take $I = X$. Then (2.3) gives

$$\left|\Pr_\ell[A_\theta] - \Pr[A_\theta]\right| \le 4(e\alpha)^{\ell-1} = e^{-\Omega(\ell)}.$$

For the larger intermediate range $\alpha(\ell-1) < \mu_\theta \le R\ell$, where $R > 2$ is a fixed constant, choose a subset $X'_\theta \subseteq X$ of size $b_\theta := \left\lfloor \frac{\alpha(\ell-1)}{p_\theta} \right\rfloor$. Then $b_\theta p_\theta = \Theta(\ell)$, while $A_\theta \subseteq A_\theta(X'_\theta)$. Consequently,

$$\Pr_\ell[A_\theta] \le \Pr_\ell[A_\theta(X'_\theta)] \le (1-p_\theta)^{b_\theta} + 4\left(\frac{eb_\theta p_\theta}{\ell-1}\right)^{\ell-1} = e^{-\Omega(\ell)}.$$

Since also $\Pr[A_\theta] = (1-p_\theta)^s \le e^{-\mu_\theta} = e^{-\Omega(\ell)}$, we obtain

$$\left|\Pr_\ell[A_\theta] - \Pr[A_\theta]\right| = e^{-\Omega(\ell)}$$

throughout this intermediate range.

For the remaining thresholds with $\mu_\theta > R\ell$, one may apply the tail inequality using the full moment order ℓ:

$$\Pr_\ell[A_\theta],\ \Pr[A_\theta] \le 8\left(\frac{2\ell}{\mu_\theta}\right)^{\ell/2}.$$

The same dyadic decomposition as above then gives a convergent geometric series, since $\ell/2 > k$.

Together with the same weighted summation as in the main argument, this yields an alternative proof: Below the scale $R\ell$, one either applies Bonferroni directly to X or to a suitably chosen subset X'_θ; above that scale, the full-order tail bound applies.

2.2 Lower Bound

We now prove that any degree of independence that universally guarantees k-min-wise hashing with multiplicative error δ must be at least $\Omega(k + \log 1/\delta)$, matching the upper bound showed before.

We first prove the $\Omega(k)$ lower bound. Fix arbitrary disjoint sets X, Y with $|X| = s$ and $|Y| = k$, where s is any positive integer. For convenience, let the hash values lie in $[0, 1)$. The elements of X are hashed independently and uniformly. For Y, first sample the hash values of $k-1$ elements, and then sample the last one so that the number of elements of Y landing in the upper half $[1/2, 1)$ is even. Once the half containing an element has been determined, its position within that half is uniform. The hash values on X and Y are mutually independent.

This distribution is exactly $(k-1)$-wise independent: the parity constraint affects only the joint distribution of all k elements, while every collection of at most $k-1$ elements remains independent and uniform. This method of imposing a parity constraint on the numbers of elements landing in the two halves was also used by [PT16] in their lower bound for min-wise hashing.

Consider the event

$$E = \{\max h(Y) < \min h(X)\},$$

and let $\mathcal{P}$ denote the event that an even number of elements of Y land in the upper half. The distribution above is precisely the fully random distribution conditioned on $\mathcal{P}$, and $\Pr[\mathcal{P}] = 1/2$. Under fully random hashing,

$$\Pr[E] = \binom{s+k}{k}^{-1}.$$

Let $N = s+k$, and let Z be the number of elements of $X \cup Y$ whose hash values lie in the lower half $[0, 1/2)$. A fully random hash function is equivalently generated in two steps: first sample N independent uniform values and sort them; then assign the N elements of $X \cup Y$ to the resulting positions according to a uniformly random permutation. Since the hash values are independent and identically distributed, this random permutation is independent of the sorted values.

The event E only requires that the first k positions be occupied by the elements of Y, whereas Z depends only on how many of the sorted values are below $1/2$. Conditioning on E therefore does not change the distribution of Z, and

$$Z \mid E \sim \mathrm{Bin}\left(N, \frac{1}{2}\right),$$

where $\mathrm{Bin}(\cdot, \cdot)$ denotes the binomial distribution.

Conditioned on E, if $Z = j < k$, then exactly $k - j$ elements of Y lie in the upper half; if $Z \geq k$, then all elements of Y lie in the lower half. It follows that

$$\begin{aligned}
\Pr[\mathcal{P} \mid E] &= \sum_{j=k-2,k-4,\ldots} \Pr[Z = j] + \sum_{j=k}^{N} \Pr[Z = j] \\
&= \frac{1}{2^N}\left(\sum_{j=k-2,k-4,\ldots} \binom{N}{j} + \sum_{j=k}^{N} \binom{N}{j}\right) \\
&= \frac{1}{2}\left(1 + \frac{1}{2^{N-1}} \sum_{j=k}^{N-1} \binom{N-1}{j}\right) \\
&= \frac{1}{2}\left(1 + \Pr\left[\mathrm{Bin}\left(N-1, \frac{1}{2}\right) \geq k\right]\right).
\end{aligned}$$

Since the $(k-1)$-wise independent distribution above is the fully random distribution conditioned on $\mathcal{P}$, Bayes' rule gives

$$\Pr[E \mid \mathcal{P}] = \frac{\Pr[E]\Pr[\mathcal{P} \mid E]}{\Pr[\mathcal{P}]} = \left(1 + \Pr\left[\mathrm{Bin}\left(s+k-1, \frac{1}{2}\right) \geq k\right]\right)\binom{s+k}{k}^{-1}.$$

Thus the multiplicative error of this $(k-1)$-wise independent distribution is exactly

$$\varepsilon_{k,s} = \Pr\left[\mathrm{Bin}\left(s+k-1, \frac{1}{2}\right) \geq k\right].$$

This formula holds for every s. If $s \geq k$, then $\varepsilon_{k,s} \geq 1/2$, while $\varepsilon_{k,s} \to 1$ as $s \to \infty$. Consequently, for every $\delta < 1$, one can choose X sufficiently large that $\varepsilon_{k,s} > \delta$. Hence $(k-1)$-wise independence does not suffice to guarantee the k-min-wise property with error δ, proving an $\Omega(k)$ lower bound.

The construction can also be discretized over a sufficiently large even range $[M]$: first determine whether each element lies in the lower or upper half of the range, and then sample it uniformly within the selected half. As M grows, the probability of the strict-ordering event approaches that in the continuous construction, so the lower bound continues to hold.

On the other hand, for the $\Omega(\log 1/\delta)$ lower bound, we notice that the definition of k-min-wise hashing includes ordinary min-wise hashing as the case $|Y| = 1$. Hence we can use the following lower bound of Pătraşcu and Thorup [PT16] for ordinary min-wise hashing.

Theorem 2.5 ([PT16, Theorem 3.1])**.** *For every set X of N keys and another key y, there exists a t-wise independent hash family*

$$\mathcal{H} = \{h : X \cup \{y\} \to [0, 1)\}$$

such that

$$\Pr_{h \sim \mathcal{H}}[h(y) < \min h(X)] = \frac{1 + 2^{-O(t)}}{N + 1}.$$

This shows that achieving multiplicative error δ requires $t = \Omega(\log 1/\delta)$. Combining this with the $\Omega(k)$ lower bound gives

$$t = \Omega\left(\max\left\{k, \log\frac{1}{\delta}\right\}\right) = \Omega\left(k + \log\frac{1}{\delta}\right).$$

Equivalently, when $\delta < 2^{-k}$, the term $\log 1/\delta$ dominates, whereas when $\delta \geq 2^{-k}$, the term k dominates.

3 Random Affine Functions

We next examine a concrete bounded-independent family. For the classical one-dimensional modular affine family

$$h(x) = (ax + b) \bmod p,$$

[BCFM98] and [PT16] showed that the multiplicative error for min-wise hashing can be $\Omega(\log N)$. We consider its natural higher-dimensional analogue over $\mathbb{F}_2$,

$$h(x) = Ax \oplus b,$$

where A and b are chosen uniformly. Although this family is pairwise independent, we show that it still incurs multiplicative error $\Omega(\log N)$ for min-wise hashing. Consequently, it also fails to satisfy the stronger k-min-wise guarantee.

Throughout this section, vectors in $\mathbb{F}_2^\ell$ are ordered lexicographically, with the first coordinate being the most significant bit. For every nonzero $y \in \mathbb{F}_2^\ell$, let $\mathsf{msb}(y)$ denote the index of its first nonzero coordinate.

Theorem 3.1. *Let $h(x) = Ax \oplus b$ be a random affine function from $\mathbb{F}_2^u$ to $\mathbb{F}_2^\ell$, where $\ell = 2u$, and where the matrix $A \in \mathbb{F}_2^{\ell \times u}$ and the vector $b \in \mathbb{F}_2^\ell$ are chosen uniformly and independently. For every integer $1 \leq m \leq u - 1$, there exists a set $S \subseteq \mathbb{F}_2^u \setminus \{0\}$ of size $N = 2^m$ such that*

$$\Pr_{A,b}[h(0) < \min h(S)] \geq (1 - 2^{-u})\frac{\log N + 1}{4N - 2} = \Theta\left(\frac{\log N}{N}\right).$$

Proof. Let $V \subseteq \mathbb{F}_2^u$ be an $(m+1)$-dimensional subspace, and fix a nonzero linear functional $\varphi \in V^*$, where V^* denotes the dual space of V. Define

$$S := \{x \in V : \varphi(x) = 1\}.$$

Since φ is nonzero, S is an affine hyperplane in V. Hence $|S| = 2^m = N$, and $0 \notin S$.

Let $\mathcal{E}$ denote the event that the restriction $A|_V$ is injective. For every nonzero vector $x \in V$, we have $\Pr[Ax = 0] = 2^{-\ell} = 2^{-2u}$. Therefore, by the union bound and the assumption $m + 1 \leq u$,

$$\Pr[\mathcal{E}] \geq 1 - (2^{m+1} - 1)2^{-2u} \geq 1 - 2^{-u}.$$

We now analyze the distribution conditioned on the event $\mathcal{E}$. Let $U := A(V)$, and write $L := A|_V : V \to U$. Then L is a linear isomorphism, while $\mathcal{Y} := A(S)$ is an affine hyperplane in U that does not contain 0.

Furthermore, conditioned on $\mathcal{E}$ and on the image space U, the map L is uniformly distributed over all linear isomorphisms from V to U. Therefore, the nonzero linear functional

$$\psi := \varphi \circ L^{-1} \in U^*$$

is uniformly distributed over $U^* \setminus \{0\}$, and $\mathcal{Y} = \{y \in U : \psi(y) = 1\}$.

Now fix A and consider the random shift b. For every nonzero vector $y \in \mathbb{F}_2^\ell$, the vectors b and $b \oplus y$ first differ at coordinate $\mathsf{msb}(y)$, and hence

$$b < b \oplus y \quad \Longleftrightarrow \quad b_{\mathsf{msb}(y)} = 0.$$

Let $\mathsf{msb}(\mathcal{Y}) := \{\mathsf{msb}(y) : y \in \mathcal{Y}\}$. Since the coordinates of b are mutually independent and uniform,

$$\Pr_b[h(0) < \min h(S) \mid A] = \Pr_b[b < b \oplus y,\ \forall y \in \mathcal{Y} \mid A] = \Pr_b[b_{\mathsf{msb}(y)} = 0,\ \forall y \in \mathcal{Y} \mid A] = 2^{-|\mathsf{msb}(\mathcal{Y})|}.$$

Let $e_1, \ldots, e_{m+1}$ be the reduced echelon basis of U, uniquely determined by U, satisfying

$$\mathsf{msb}(e_1) < \cdots < \mathsf{msb}(e_{m+1}).$$

For every $y = \sum_{i=1}^{m+1} \alpha_i e_i$, if t is the smallest index such that $\alpha_t = 1$, then $\mathsf{msb}(y) = \mathsf{msb}(e_t)$.

With respect to this basis, write ψ as

$$\psi\left(\sum_{i=1}^{m+1} \alpha_i e_i\right) = \sum_{i=1}^{m+1} c_i \alpha_i.$$

Since ψ is uniformly distributed over $U^* \setminus \{0\}$, the coefficient vector $c = (c_1, \ldots, c_{m+1})$ is uniformly distributed over $\mathbb{F}_2^{m+1} \setminus \{0\}$. Moreover,

$$\mathcal{Y} = \left\{\sum_{i=1}^{m+1} \alpha_i e_i : \sum_{i=1}^{m+1} c_i \alpha_i = 1\right\}.$$

Fix $t \in [m+1]$. The index $\mathsf{msb}(e_t)$ belongs to $\mathsf{msb}(\mathcal{Y})$ if and only if there exists a coordinate vector satisfying $\alpha_1 = \cdots = \alpha_{t-1} = 0$ and $\alpha_t = 1$ such that

$$c_t + \sum_{i=t+1}^{m+1} c_i \alpha_i = 1.$$

This equation is solvable if and only if $c_t, \dots, c_{m+1}$ are not all zero.

Let $T := \max\{i : c_i = 1\}$. The preceding condition is equivalent to $t \le T$, and hence $|\mathsf{msb}(\mathcal{Y})| = T$. Since c is uniformly distributed over all nonzero vectors,

$$\Pr[T = j \mid \mathcal{E}] = \frac{2^{j-1}}{2^{m+1}-1}, \qquad j \in [m+1].$$

Therefore,

$$\mathbb{E}[2^{-|\mathsf{msb}(\mathcal{Y})|} \mid \mathcal{E}] = \sum_{j=1}^{m+1} \frac{2^{j-1}}{2^{m+1}-1} 2^{-j} = \frac{m+1}{2(2^{m+1}-1)}.$$

Finally,

$$\Pr[h(0) < \min h(S)] \ge \Pr[\mathcal{E}] \cdot \mathbb{E}[2^{-|\mathsf{msb}(\mathcal{Y})|} \mid \mathcal{E}] \ge (1-2^{-u}) \frac{m+1}{2(2^{m+1}-1)} = (1-2^{-u}) \frac{\log N + 1}{4N-2}.$$

□

4 Simple Tabulation Hashing

Simple tabulation hashing is one of the standard constructions used for pairwise-independent hashing; in fact, it is 3-wise independent. Pătraşcu and Thorup [PT12] showed that, despite its limited independence, simple tabulation provides strong guarantees for ordinary min-wise hashing. We show that this positive behavior does not extend to k-min-wise hashing. More precisely, for every $k \ge 4$, simple tabulation incurs multiplicative error $\Omega(N)$. Our result can therefore be viewed as a negative extension of their analysis from ordinary min-wise hashing to its k-min-wise generalization.

Recall that a simple tabulation hash function $h : \Sigma^c \to \{0,1\}^w$ is defined by

$$h(x_1, \dots, x_c) = T_1[x_1] \oplus \cdots \oplus T_c[x_c],$$

where all table entries $T_i[a]$ are mutually independent and uniformly distributed in $\{0,1\}^w$. We identify a hash value with the corresponding w-bit binary fraction in $[0,1)$. A pair $\alpha = (i,a) \in [c] \times \Sigma$ is called a *position-character*, and we write $T[\alpha] := T_i[a]$. For a key $x = (x_1, \dots, x_c)$, let

$$P(x) := \{(i, x_i) : i \in [c]\}$$

denote the set of its position-characters.

Theorem 4.1. *Let $c \ge 2$ and let the hash range have size $M = 2^w$. For every positive integer N satisfying*

$$M \ge 2N \qquad \text{and} \qquad (|\Sigma| - 2)|\Sigma|^{c-1} \ge N,$$

there exist disjoint sets $X, Y \subseteq \Sigma^c$ with $|X| = N$ and $|Y| = 4$ such that

$$\Pr\left[\max h(Y) < \min h(X)\right] \ge \frac{1}{128 N^3}.$$

Consequently,

$$\frac{\Pr\left[\max h(Y) < \min h(X)\right]}{\binom{N+4}{4}^{-1}} > \frac{N}{3072}.$$

In particular, simple tabulation hashing incurs multiplicative error $\Omega(N)$ for 4-min-wise hashing.

The proof consists of two lemmas. The first exploits the xor dependency among four keys forming a combinatorial rectangle. The second uses an ordered exposure of the position-characters to control the hash values of the remaining keys.

Lemma 4.2. *Fix two distinct characters $0, 1 \in \Sigma$, and define*

$$y_{ab} := (a, b, 0, \ldots, 0), \qquad a, b \in \{0, 1\}.$$

Let $Y := \{y_{00}, y_{01}, y_{10}, y_{11}\}$. For every dyadic threshold $p = 2^{-r}$ with $r \leq w$,

$$\Pr[\max h(Y) < p] = p^3.$$

Proof. Write $H_{ab} := h(y_{ab})$. Every position-character appearing among the four keys occurs an even number of times. Therefore,

$$H_{00} \oplus H_{01} \oplus H_{10} \oplus H_{11} = 0,$$

and hence

$$H_{11} = H_{00} \oplus H_{01} \oplus H_{10}.$$

Moreover, H_{00}, H_{01}, H_{10} are independent and uniformly distributed, since simple tabulation is 3-wise independent.

The event $H_{ab} < p$ means precisely that the first r bits of H_{ab} are all zero. Thus, if $H_{00}, H_{01}, H_{10} < p$, then the first r bits of their xor are also zero, and hence $H_{11} < p$. Consequently,

$$\{\max h(Y) < p\} = \{H_{00} < p,\ H_{01} < p,\ H_{10} < p\}.$$

Since H_{00}, H_{01}, H_{10} are independent and uniform,

$$\Pr[\max h(Y) < p] = p^3.$$

□

Lemma 4.3. *Let $Y \subseteq \Sigma^c$, and let*

$$Q := \bigcup_{y \in Y} P(y)$$

be the set of position-characters appearing in Y. Let $X \subseteq \Sigma^c$ be a set of N keys such that $P(x) \nsubseteq Q$ for every $x \in X$. Then, for every dyadic threshold $p = 2^{-r}$ satisfying $r \leq w$ and $pN \leq 1$,

$$\Pr[\min h(X) \geq p \mid \max h(Y) < p] \geq 1 - pN.$$

Proof. The table values $T[\alpha]$ associated with distinct position-characters $\alpha \in [c] \times \Sigma$ are mutually independent. We may therefore expose them one at a time in any fixed order without changing their joint distribution.

Choose a total order $\prec$ on $[c] \times \Sigma$ in which every position-character in Q precedes every position-character outside Q. For each $\alpha \notin Q$, define

$$G_\alpha := \{x \in X : \alpha = \max\nolimits_\prec P(x)\}.$$

Since $P(x) \nsubseteq Q$ for every $x \in X$, each key contains a position-character outside Q. Hence its $\prec$-maximum lies outside Q, and the nonempty groups G_α form a partition of X:

$$X = \bigsqcup_{\alpha \notin Q} G_\alpha, \qquad \sum_{\alpha \notin Q} |G_\alpha| = N.$$

The event $\{\max h(Y) < p\}$ depends only on the table entries indexed by Q. Thus, after conditioning on this event, the table entries indexed outside Q remain mutually independent and uniform. We first expose the entries indexed by Q and then expose the remaining entries according to $\prec$.

For a position-character α, let $h(\prec \alpha)$ denote all table values exposed before $T[\alpha]$. Consider a nonempty group G_α. For every $x \in G_\alpha$, all position-characters of x other than α precede α, and hence

$$h(x) = T[\alpha] \oplus a_x := T[\alpha] \oplus \bigoplus_{\beta \in P(x), \beta \prec \alpha} T[\beta],$$

where a_x is determined by $h(\prec \alpha)$.

Moreover, if $\beta \prec \alpha$ and $x \in G_\beta$, then every position-character of x is at most β under $\prec$. Therefore, every previously processed group depends only on table entries contained in $h(\prec \alpha)$. In particular, even after conditioning on all previous groups avoiding $[0, p)$, the value $T[\alpha]$ remains fresh and uniform.

Consequently, for every fixed $x \in G_\alpha$,

$$\Pr[h(x) < p \mid h(\prec \alpha)] = p.$$

The hash values within G_α need not be independent, but the union bound gives

$$\Pr\left[h(G_\alpha) \cap [0, p) = \varnothing \mid h(\prec \alpha)\right] \geq 1 - p|G_\alpha|.$$

Applying this bound successively to all nonempty groups yields

$$\Pr[\min h(X) \geq p \mid \max h(Y) < p] \geq \prod_{\alpha \notin Q} \big(1 - p|G_\alpha|\big) \geq 1 - p \sum_{\alpha \notin Q} |G_\alpha| = 1 - pN.$$

□

We are now ready to combine the two lemmas to prove the main theorem.

Proof of Theorem 4.1. Fix two distinct characters $0, 1 \in \Sigma$ and let

$$Y := \{(a, b, 0, \ldots, 0) : a, b \in \{0, 1\}\}.$$

Choose any set X of N distinct keys whose first character lies outside $\{0, 1\}$. Such a set exists because $(|\Sigma| - 2)|\Sigma|^{c-1} \geq N$. The sets X and Y are disjoint.

Let

$$Q := \bigcup_{y \in Y} P(y).$$

For every $x \in X$, the first position-character $(1, x_1)$ does not belong to Q. Hence $P(x) \nsubseteq Q$, and Lemma 4.3 applies.

Set

$$r := \lceil \log_2(2N) \rceil \qquad \text{and} \qquad p := 2^{-r}.$$

Since $M = 2^w \geq 2N$, we have $r \leq w$. Moreover,

$$\frac{1}{4N} < p \leq \frac{1}{2N}.$$

By Lemma 4.2,

$$\Pr[\max h(Y) < p] = p^3.$$

By Lemma 4.3,

$$\Pr[\min h(X) \geq p \mid \max h(Y) < p] \geq 1 - pN \geq \frac{1}{2}.$$

The event $\{\max h(Y) < p\} \cap \{\min h(X) \geq p\}$ implies $\{\max h(Y) < \min h(X)\}$. Therefore,

$$\begin{aligned}
\Pr[\max h(Y) < \min h(X)] &\geq \Pr[\max h(Y) < p,\ \min h(X) \geq p] \\
&= \Pr[\max h(Y) < p] \cdot \Pr[\min h(X) \geq p \mid \max h(Y) < p] \\
&\geq p^3 \cdot \frac{1}{2} \geq \frac{1}{128N^3}.
\end{aligned}$$

Under a uniformly random ordering of $X \cup Y$, the probability that the four prescribed elements of Y precede all elements of X is $\binom{N+4}{4}^{-1}$. Hence

$$\frac{\Pr[\max h(Y) < \min h(X)]}{\binom{N+4}{4}^{-1}} \geq \frac{\binom{N+4}{4}}{128N^3} > \frac{N}{3072}.$$

The ratio grows linearly with N, proving the claim. $\square$

Remark 4.4. *The obstruction above inherently uses four prescribed keys. For $|Y| \leq 3$, the argument of [PT12] can be adapted to obtain analogous positive guarantees. We do not pursue this direction here.*

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A Lower Bounds for Seed Length

Proposition A.1. *Let $1 \le k \le N/2$ and $0 < \delta < 1$, and let $\mathcal{H}$ be a finite k-min-wise hash family with multiplicative error δ. Then*

$$|\mathcal{H}| \ge \max\left\{\binom{N}{k}, \min\left\{\frac{1}{\delta}, \mathsf{lcm}(1,\dots,N)\right\}\right\}.$$

Consequently, the seed length of $\mathcal{H}$ is

$$\Omega\left(k \log \frac{N}{k} + \min\left\{\log\frac{1}{\delta}, N\right\}\right).$$

In particular, when $k \le N^{1-c_1}$ and $\delta = N^{-c_2}$, the seed length is $\Omega(k \log N)$.

Proof. Let $D := |\mathcal{H}|$. We first prove $D \ge \binom{N}{k}$. Fix any set $Y \subseteq [N]$ of size k, and apply the k-min-wise guarantee with $X = [N] \setminus Y$. Since $\delta < 1$,

$$\Pr_{h \sim \mathcal{H}}[\max h(Y) < \min h([N] \setminus Y)] \ge (1-\delta)\binom{N}{k}^{-1} > 0.$$

Hence some function in $\mathcal{H}$ realizes Y as the strict bottom-k set of $[N]$. A fixed hash function can realize at most one set as its strict bottom-k set. Since every one of the $\binom{N}{k}$ possible choices of Y must be realized, we obtain $D \ge \binom{N}{k}$.

We next derive a lower bound depending on δ. Fix any integer $m \in [N]$, any set $S \subseteq [N]$ of size m, and any element $y \in S$. Since the k-min-wise property includes ordinary min-wise hashing and $\mathcal{H}$ is a finite family, there exists an integer a such that

$$\frac{a}{D} = \Pr_{h \sim \mathcal{H}}[h(y) < \min h(S \setminus \{y\})] = (1 \pm \delta)\frac{1}{m}.$$

Therefore,

$$|ma - D| \le \delta D.$$

Suppose that $D < 1/\delta$. Then $\delta D < 1$. Since $ma - D$ is an integer, the preceding inequality forces $ma - D = 0$, and hence $m \mid D$. As this holds for every $m \in [N]$, we have $\mathsf{lcm}(1, \dots, N) \mid D$. Thus either $D \ge 1/\delta$, or $D \ge \mathsf{lcm}(1, \dots, N)$, and consequently

$$D \ge \min\left\{\frac{1}{\delta}, \mathsf{lcm}(1, \dots, N)\right\}.$$

Combining the two lower bounds proves the claimed bound on $|\mathcal{H}|$. Taking logarithms and using $\log \binom{N}{k} = \Omega(k \log(N/k))$ for $k \le N/2$ and $\log \mathsf{lcm}(1, \dots, N) = \Theta(N)$ (see, e.g., [San21]) gives

$$\log D = \Omega\left(k \log \frac{N}{k} + \min\left\{\log \frac{1}{\delta}, N\right\}\right).$$

$\square$